Retrieval-Augmented Generation (RAG)

Connecting language models to a document knowledge base

Purpose

This short reference note is written for learning, semantic retrieval, and grounded question-answering experiments. It uses original explanatory content and is organized into coherent sections rather than unrelated filler.

1. The RAG Idea

Retrieval-augmented generation combines information retrieval with language-model generation. Instead of relying only on model parameters, the application searches an external collection for passages relevant to a question. Those passages are inserted into the prompt, and the model generates an answer using the supplied evidence. RAG is useful for private documents, frequently changing knowledge, and applications that need traceable sources.

2. A Typical Pipeline

A document-ingestion pipeline accepts files, extracts text, cleans it, splits it into chunks, creates embeddings, and stores vectors alongside text and metadata. At query time, the system embeds the question, searches the vector store, selects the top-k candidates, builds a prompt containing the evidence, and calls the language model. The application then returns an answer and source details. Each stage can introduce a distinct failure mode.

3. Why Chunking Matters

A whole book may be too large to embed or retrieve as one unit, so documents are divided into chunks. Tiny chunks can lose definitions and context; very large chunks can mix topics and make matching less precise. Overlap can preserve information around boundaries, though too much overlap creates duplicated content and increases storage. The best strategy depends on document structure and question types, so it should be evaluated empirically.

4. Metadata and Source Traceability

Each chunk should retain metadata such as document ID, file name, page number, section title, and chunk index where available. Metadata filters can restrict retrieval to a specific file or category. Source details also let a user verify an answer and let a developer determine whether an error came from extraction, retrieval, or generation. Keep stable identifiers across ingestion runs when practical, or explicitly record when they change.

5. Retrieval and Generation Are Different

Retrieval selects candidate evidence; generation turns that evidence into prose. Good generation cannot reliably repair missing evidence, and accurate retrieval does not guarantee a correct final answer. Measure retrieval quality independently from answer quality. For example, record whether a relevant chunk appears in the top three results, then separately assess whether the generated response is correct and supported.

6. RAG Versus Fine-Tuning

RAG makes external information available at inference time without updating the model's parameters. Fine-tuning changes model behavior through additional training. RAG is often a better first choice for a document-search assistant because sources can be updated by re-ingesting documents. Fine-tuning may be useful for specialized response formats or repeated behaviors, but it is not a substitute for retrieving up-to-date source material.

Key Takeaways

RAG has two major phases: ingestion and question answering. Quality depends on every stage, including text extraction, chunking, embeddings, search, prompt construction, and grounded generation.

End of document • Use source passages as evidence and verify claims during evaluation.